Clinic Management System

Milestone 3 — Dataset Preprocessing

Database Lab | Version Control Submission

1. Overview

This document presents the complete Milestone 3 submission for the Clinic Management System database project. It covers synthetic data generation for all five tables, a data preprocessing explanation, a project-specific dataflow description, and a list of exported CSV files ready for database loading in Milestone 5.

The five tables in this project are: Department, Patient, Doctor, Appointment, and Prescription. All data has been generated to be realistic, consistent, and fully compliant with the foreign key relationships established in Milestone 1 and the normalized schema documented in Milestone 2.

2. Schema & Relationships

Before generating data, the table relationships were confirmed to ensure foreign key integrity during generation:

Table	Primary Key	Foreign Keys	Depends On
Department	department_id	—	None (root table)
Patient	patient_id	—	None (root table)
Doctor	doctor_id	department_id → Department	Department
Appointment	appointment_id	patient_id → Patient, doctor_id → Doctor	Patient, Doctor
Prescription	prescription_id	appointment_id → Appointment, doctor_id → Doctor, patient_id → Patient	Appointment

3. Synthetic Data Generation

Since this project does not use a real-world dataset, all data was generated synthetically using Python 3. The generation logic was written manually to produce realistic, Pakistan-locale data with proper foreign key linkages. No random placeholder text was used.

3.1 Tool & Approach

Tool Used	Python 3 (standard library: csv, random, datetime)
Locale	Pakistan — Pakistani names, cities, phone formats, clinic names
Random Seed	Fixed seed (42) for reproducibility
Generation Order	Department → Patient → Doctor → Appointment → Prescription

3.2 Department Table

Rows Generated	6
Description	Six real clinical departments found in a typical Pakistani clinic
Columns	department_id, department_name, location, phone

Sample Data (first 5 rows):

department_id	department_name	location	phone
1	General Medicine	Block A, Ground Floor	051-1110001
2	Cardiology	Block B, First Floor	051-1110002
3	Dermatology	Block A, First Floor	051-1110003
4	Orthopedics	Block C, Ground Floor	051-1110004
5	Gynecology	Block D, Second Floor	051-1110005

3.3 Patient Table

Rows Generated	60
Description	Patients with Pakistani names, cities, realistic date-of-birth, blood group, and contact information
Columns	patient_id, first_name, last_name, date_of_birth, gender, phone, email, address, blood_group, registration_date
Phone Format	03XX-XXXXXXX (Pakistani mobile format)
Date of Birth Range	1955 – 2005
Registration Date Range	2022 – 2024

Sample Data (first 5 rows):

patient_id	first_name	last_name	DOB	gender	phone	email	blood_group	reg_date
1	Ali	Siddiqui	1970-04-05	Male	0342458591	ali.siddiqui1@gmail.com	A+	2022-02-07
2	Sohail	Baig	1956-09-07	Male	0318536477	sohail.baig2@gmail.com	A+	2022-12-14
3	Maryam	Qureshi	1989-03-22	Female	0311724831	maryam.qureshi3@gmail.com	B+	2023-01-18
4	Hamza	Akhtar	2001-07-14	Male	0304193826	hamza.akhtar4@gmail.com	O+	2023-03-09
5	Fatima	Rehman	1995-11-30	Female	0321837461	fatima.rehman5@gmail.com	AB+	2022-08-25

3.4 Doctor Table

Rows Generated	18 (3 doctors per department)
Description	Doctors assigned to specific departments with realistic specializations and consultation fees
Columns	doctor_id, first_name, last_name, specialization, phone, email, department_id, consultation_fee, hire_date
Specializations	General Physician, Cardiologist, Dermatologist, Orthopedic Surgeon, Gynecologist, Pediatrician
Fee Range	Rs. 800 – Rs. 2,400 depending on specialization
Hire Date Range	2010 – 2020

Sample Data (first 5 rows):

doctor_id	first_name	last_name	specialization	department_id	fee	hire_date
1	Imran	Mahmood	General Physician	1	1200	2019-10-08
2	Khalid	Farooqi	General Physician	1	1000	2017-05-28
3	Rubina	Chaudhry	General Physician	1	800	2014-02-11
4	Yasir	Siddiqui	Cardiologist	2	2200	2016-07-03
5	Naveed	Baig	Cardiologist	2	2000	2018-09-15

3.5 Appointment Table

Rows Generated	120
Description	Appointments between existing patients and doctors, spread across 2023, with realistic statuses and clinical notes
Columns	appointment_id, patient_id, doctor_id, appointment_date, appointment_time, status, notes
Status Values	Completed (approx. 65%), Scheduled, Cancelled, No-Show
Date Range	January 2023 – December 2023
Time Range	08:00 – 16:45 in 15-minute slots
FK Integrity	patient_id and doctor_id are always valid existing IDs
No Conflicts	No two appointments share the same doctor, date, and time slot

Sample Data (5 rows — mix of statuses):

appt_id	patient_id	doctor_id	date	time	status	notes
3	12	8	2023-09-17	10:00:00	Completed	Skin rash on arms since last week.
4	40	16	2023-11-15	09:15:00	Completed	Child vaccination and growth checkup.
5	60	10	2023-08-23	11:30:00	Completed	Joint pain in left knee for two months.
6	25	4	2023-04-11	14:00:00	Scheduled	Routine checkup requested by patient.
7	18	7	2023-07-22	08:30:00	Cancelled	Patient reports chest pain.

3.6 Prescription Table

Rows Generated	70
Description	Prescriptions created only for Completed appointments. Each prescription is linked to the appointment, doctor, and patient of that visit.
Columns	prescription_id, appointment_id, doctor_id, patient_id, prescribed_date, medicine_name, dosage, duration, notes
Medicine Selection	Medicines are matched to the doctor's specialization (e.g., Cardiologist prescribes Aspirin, Atenolol)
Dosage Values	Once daily, Twice daily, Three times daily, As needed, Before meals, After meals

Duration Values	3 days, 5 days, 7 days, 10 days, 14 days, 1 month
FK Integrity	appointment_id always references a Completed appointment; doctor_id and patient_id match that appointment's records

Sample Data (first 5 rows):

presc_id	appt_id	doctor_id	patient_id	date	medicine_name	dosage	duration
1	4	16	40	2023-11-15	Zinc Syrup 10mg	As needed	3 days
2	5	10	60	2023-08-23	Calcium Carbonate 500mg	Before meals	7 days
3	3	8	12	2023-09-17	Clindamycin Gel 1%	Twice daily	14 days
4	9	2	33	2023-03-05	Aspirin 75mg	Once daily	1 month
5	11	5	47	2023-06-28	Folic Acid 5mg	Once daily	1 month

4. Data Preprocessing & Cleaning

Even though the data was synthetically generated, a preprocessing checklist was applied during generation and verified after export. The following steps were taken to ensure the data is clean and ready for database loading:

4.1 Duplicate Removal

Duplicates were prevented at the generation stage using the following checks:

- Patient: Each patient_id is unique (auto-incremented). No two patients share the same combination of name, date of birth, and phone number.
- Doctor: Each doctor_id is unique. No two doctors share the same email address.
- Appointment: A time-slot conflict check was applied during generation — no two appointments share the same doctor_id, appointment_date, and appointment_time. This prevents double-booking.
- Prescription: Each appointment_id appears only once in the Prescription table. One prescription is issued per completed appointment.

4.2 Handling Null Values

All columns marked NOT NULL in the schema were guaranteed to have a value before writing to CSV:

- Required fields (first_name, last_name, date_of_birth, gender, phone, appointment_date, appointment_time, status, medicine_name, dosage, duration) are always populated.
- Optional fields (notes in Appointment, notes in Prescription) are allowed to be empty strings. They are written as empty fields in the CSV rather than NULL to avoid import errors.
- No column in any CSV row is missing or left with a placeholder such as 'N/A' or 'TBD'.

4.3 Formatting Consistency

All date and phone values follow a consistent format throughout all five CSV files:

Field	Format Used	Example
Date columns	YYYY-MM-DD	2023-09-17
Time columns	HH:MM:SS (24-hour)	14:30:00
Phone numbers	Pakistani mobile (10 digits, starts with 03)	0312847619
Email addresses	lowercase, no spaces	dr.imran.mahmood@clinic.pk
Gender	Fixed values: Male / Female	Female
Status	Fixed values: Completed / Scheduled / Cancelled / No-Show	Completed

4.4 Data Type Consistency

Each column in the CSV files matches the data type expected by the corresponding database column:

- Integer columns (IDs, consultation_fee) contain only numeric values with no currency symbols or text.
- Decimal values (consultation_fee) are stored as plain numbers, e.g., 1200 not Rs.1,200.
- Date columns contain only valid calendar dates in YYYY-MM-DD format — no dates like '31st March' or '03/31/23'.
- String columns contain no leading or trailing whitespace.

4.5 Foreign Key Integrity

Foreign key relationships were enforced during the data generation process itself:

- Every doctor.department_id value exists in department.department_id.
- Every appointment.patient_id value exists in patient.patient_id.
- Every appointment.doctor_id value exists in doctor.doctor_id.
- Every prescription.appointment_id references only appointments with status = 'Completed'.
- Every prescription.doctor_id and prescription.patient_id match the doctor and patient recorded in the referenced appointment row.

- No orphan records exist in any table.

5. Dataflow Description

This section describes how data enters, moves through, and exits the Clinic Management System. The dataflow is specific to this project and reflects the actual relationships between the five tables.

5.1 Where Data Enters the System

Data enters the system at two independent starting points:

Entry Point 1	Department table — Departments are set up first by the clinic administrator. They have no dependencies on any other table and serve as a foundation for assigning doctors.
Entry Point 2	Patient table — Patient records are created when a new patient registers at the clinic. Patient data also has no dependencies and can be entered independently.

5.2 How Data Moves Through the Tables

Once departments and patients exist, the remaining tables are populated in a defined sequence:

1. Doctor Registration: A doctor is added to the Doctor table and assigned to an existing department via department_id. This means the Department table must already have data before any doctor record can be created.
2. Appointment Booking: When a patient visits the clinic, an appointment record is inserted into the Appointment table. This record references an existing patient_id from the Patient table and an existing doctor_id from the Doctor table. Both must exist before an appointment can be recorded.
3. Prescription Issuance: After an appointment is completed, the attending doctor issues a prescription. This is recorded in the Prescription table, which references the appointment_id of the completed appointment, as well as the doctor_id and patient_id for direct access. A prescription can only be created after its linked appointment exists and is marked as Completed.

5.3 Table Dependency Order

The dependency chain from left to right shows which tables must be populated first:

Step	Table	Depends On	Can Be Populated?
1	Department	Nothing	Yes — first table to populate

Step	Table	Depends On	Can Be Populated?
2	Patient	Nothing	Yes — independently of Department
3	Doctor	Department	Only after Department has data
4	Appointment	Patient + Doctor	Only after both Patient and Doctor have data
5	Prescription	Appointment (Completed)	Only after Appointment has Completed rows

5.4 What Outputs Are Generated

The database supports the following outputs through SQL queries:

- Patient History Report: A JOIN across Patient, Appointment, and Prescription tables returns a full history of a patient's visits and prescribed medicines.
- Doctor Schedule: The Appointment table filtered by doctor_id and date range produces a daily or weekly appointment schedule for any doctor.
- Department Summary: A JOIN between Doctor and Department returns the number of doctors and their specializations per department.
- Prescription Report: A JOIN across Prescription and Appointment retrieves all medicines prescribed during a given time period, grouped by doctor or patient.
- Appointment Status Report: Aggregating the Appointment table by status (Completed, Scheduled, Cancelled, No-Show) gives clinic management a summary of appointment outcomes.

6. CSV Files

Five CSV files have been generated and exported — one per table. These files are ready to be loaded into the database in Milestone 5 using LOAD DATA INFILE or INSERT statements.

CSV File	Table	Rows	Columns
department.csv	Department	6	department_id, department_name, location, phone
patient.csv	Patient	60	patient_id, first_name, last_name, date_of_birth, gender, phone, email, address, blood_group, registration_date
doctor.csv	Doctor	18	doctor_id, first_name, last_name, specialization, phone, email, department_id, consultation_fee, hire_date
appointment.csv	Appointment	120	appointment_id, patient_id, doctor_id, appointment_date, appointment_time, status, notes
prescription.csv	Prescription	70	prescription_id, appointment_id, doctor_id, patient_id, prescribed_date, medicine_name, dosage, duration, notes

6.1 File Details

- All CSV files use comma (,) as the field delimiter.
- Text fields containing commas are wrapped in double quotes.
- The first row of each file is a header row containing column names.
- All files use UTF-8 encoding.
- Empty optional fields (e.g., blank notes) are written as empty strings between commas, not as NULL.